Beyond curl-noise: helicity and phase coherence controls for divergence-free flow fields

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Abstract

We present a solver-free method for synthesizing divergence-free vector fields for the procedural animation of smoke, particles, and other advected media, giving the artist three independent dials on top of the usual noise: the spectral slope (how ragged versus silky the flow looks), the *helicity* (whether the swirls corkscrew left or right), and the *phase coherence* (whether the motion organizes into distinct rolling vortices or stays an even haze). The construction works in the frequency domain on the *helical* basis—the natural directions in which a swirl reinforces its own rotation—which is divergence-free by construction and turns handedness into a single dial, continuous from mirror-symmetric (no preferred twist) to maximally helical (every mode fully corkscrewed—Beltrami mode by mode). The phase-coherence dial is, to our knowledge, new to procedural flow: it slides the field from formless noise to organized filaments while holding the amount of motion at every scale fixed, so the flow reorganizes without speeding up or slowing down—an axis that curl-noise does not expose. We show that making the field real-valued (the step every graphics implementation needs) does not wash out the chosen handedness, give the construction in two and three dimensions, and provide two interchangeable ways to evaluate it: an FFT bake for offline authoring and an analytic mode-sum that runs per pixel in a fragment shader (GLSL included). Solid obstacles are handled in the same framework by ramping the vector potential before the curl, which keeps the field exactly divergence-free and tangent to the surface with no solver and no pressure projection. The payoff for the artist is direct control over how a flow twists and how tightly it organizes—qualities that ordinary curl-noise leaves fixed—at no added solver cost downstream.

1 Introduction

Procedural flow fields are a workhorse of visual effects: rather than simulate a fluid, one generates a static or slowly-evolving velocity field and advects particles, smoke density, or other tracers through it. The field must be *divergence-free* so that tracers are transported without spurious compression or rarefaction. The standard construction is *curl-noise* [1]: take a vector (or scalar) noise potential [2] and use its curl, which is divergence-free identically. Curl-noise is cheap, tileable with care, and ubiquitous, and it sits within a broad family of procedural noise functions [15], some of which already expose spectral control by example [16]. It has itself been extended in recent years—boundary-respecting pointwise variants [13] and differentiable formulations [14]—but these enlarge its geometry and controllability rather than the two qualities we target below.

Its expressive range is, however, narrow. Curl-noise is isotropic and exposes essentially two knobs—frequency and amplitude. This paper adds three dials to a divergence-free field. One, the spectral slope, sharpens control the broader noise family already offers in weaker forms [15, 16] into

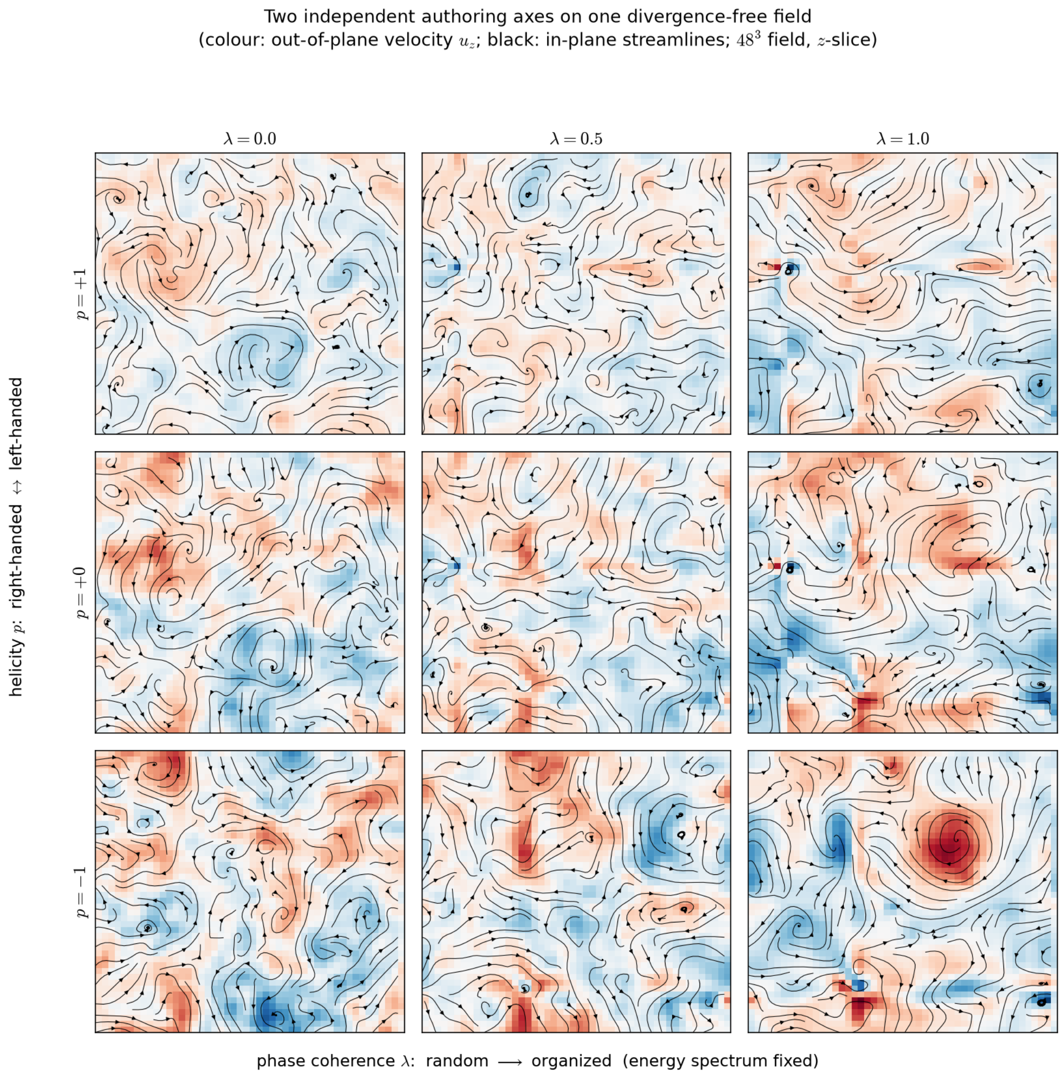

Figure 1: Two independent authoring axes on a single divergence-free field. **Left to right:** the phase-coherence knob λ (§5) carries the field from random-phase noise to organized structure *at a fixed power spectrum* (the vortices sharpen and separate as you move right). **Top to bottom:** the helicity knob p (§3) sweeps the chirality from right-handed (top) through balanced (middle) to left-handed (bottom). Chirality is a correlation, not a colour: in a helical field the in-plane swirl and the out-of-plane velocity (the colour, u_z) are locked in a definite sense, and that sense reverses between the top and bottom rows, while the middle row has no preferred twist. The two axes are orthogonal: the total energy is identical across every row (it does not depend on p at all), and the net helicity $\int \mathbf{u} \cdot \boldsymbol{\omega}$ is set by p alone—exactly antisymmetric, $+1.3 \cdot 10^{-3}$, 0, $-1.3 \cdot 10^{-3}$ for the three rows—and is unchanged by λ to machine precision. Colour is the out-of-plane velocity u_z; black curves are in-plane streamlines; each panel is a z-slice of a 48^3 field.

an exact per-scale envelope (§2). The other two are qualities that distinguish real turbulent flow and that no curl-noise variant exposes directly:

- **Helicity** (chirality): whether the swirls of the flow corkscrew left or right. Watch a dye streak carried by the field—a right-handed flow winds it into a clockwise helix, a left-handed one into its mirror, and a mirror-symmetric flow shows no preference. Quantitatively, helicity $\int \mathbf{u} \cdot \boldsymbol{\omega} \, d\mathbf{x}$ is a genuine three-dimensional invariant of ideal flow [8]; a field can range from mirror-symmetric to fully helical (a Beltrami field, $\boldsymbol{\omega} \parallel \mathbf{u}$, in which every swirl is maximally corkscrewed) [6, 9].
- **Phase organization**: whether the motion lines up into distinct rolling vortices or stays an even, structureless haze. At a *fixed* power spectrum—the same amount of motion at every scale, so the flow neither speeds up nor slows down—a field with random inter-mode phases looks featureless (Gaussian), whereas correlated phases pull that same energy into coherent filaments and sheets. That the phases, not the amplitude spectrum, carry the structural information of a field is well established in turbulence and in cosmology [10].

We build the field directly in the frequency domain from a prescribed spectral tensor, which makes both qualities into explicit controls while retaining exact divergence-freeness (Fig. 1). The contributions are:

1. a divergence-free spectral construction with a single *helicity* parameter $p \in [-1, 1]$, realized on the helical basis of Waleffe and of Craya–Herring (§2.2), and reducing in two dimensions to a stream-function curl (§2);
2. a proof that the Hermitian symmetrization required to obtain a *real* field preserves the helicity polarization—the bias is not washed out by making the field real (§4, Proposition 1);
3. a *phase-coherence* control that interpolates inter-mode phases from random to structured at fixed power spectrum (§5);
4. an analytic Beltrami mode-sum variant that evaluates per pixel with no FFT, suitable for a real-time fragment shader, together with its exact and statistical properties (§6);
5. a solid-obstacle treatment that ramps the vector potential before taking the curl, keeping the field exactly divergence-free and tangent to the surface—free-slip and no-penetration—for any obstacle with a smooth signed distance, with no solver (§7);
6. numerical validation of divergence-freeness, reality, helicity control, and the obstacle guarantees for both constructions (§8, §7).

The method is an authoring/bake tool in the spirit of spectral turbulence synthesis for graphics [3, 4]. Synthesizing divergence-free random fields with a prescribed spectrum is itself long established outside graphics—kinematic simulation sums unsteady random Fourier modes constrained to be solenoidal [19], and divergence-free synthetic-turbulence generators build the field as the curl of a random vector potential with a target spectrum [20]. Our contribution is not that machinery but the pair of *artist-facing* controls it does not expose—a single helicity/chirality dial and a phase-coherence dial at fixed spectrum—together with the in-shader mode-sum and the potential-ramp obstacle treatment. §9 positions the method against curl-noise and states its limits.

2 Divergence-free spectral fields

We work on a periodic box $\mathbb{T}^d = [0, 2\pi)^d$, $d \in \{2, 3\}$, with wavevectors $\mathbf{k} \in \mathbb{Z}^d$. A field $\mathbf{u}(\mathbf{x}) = \sum_{\mathbf{k}} \hat{\mathbf{u}}(\mathbf{k}) \, e^{i\mathbf{k}\cdot\mathbf{x}}$ is divergence-free iff $\mathbf{k} \cdot \hat{\mathbf{u}}(\mathbf{k}) = 0$ for every $\mathbf{k}$. The spectral slope enters through a radial envelope

$$E(\mathbf{k}) = |\mathbf{k}|^{-s} \, e^{-|\mathbf{k}|^2/k_c^2}, \tag{1}$$

with slope s and a smooth high-frequency cutoff k_c. Here E is the *amplitude* envelope, $|\hat{\mathbf{u}}(\mathbf{k})| \propto E(\mathbf{k})$, so the modal energy scales as $|\mathbf{k}|^{-2s}$ and the energy in a d-dimensional shell as $|\mathbf{k}|^{d-1-2s}$; we quote s as the amplitude exponent throughout. (In three dimensions the envelope is placed directly on the velocity; in the 2D stream-function path (2) it is placed on the vorticity, so the velocity there carries an extra $|\mathbf{k}|^{-1}$ and a given s means a slope steeper by one. We keep the two parametrizations because each is the natural one in its construction.) Steep s concentrates energy in a few large scales (smooth, silky flow); shallow s spreads it across scales (fine, ragged flow) (Fig. 2).

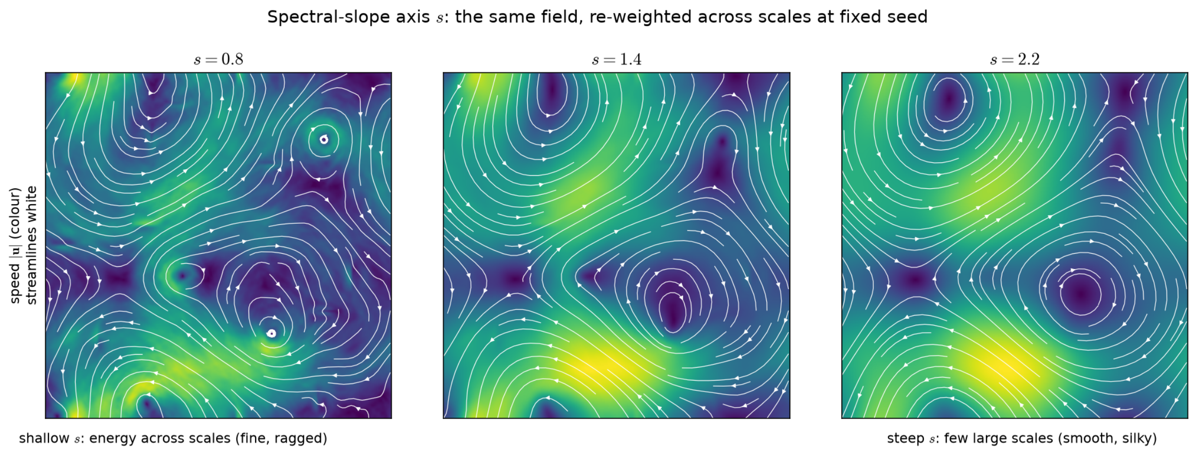

Figure 2: The spectral-slope axis s of the envelope (1), on one two-dimensional field at fixed random seed and phase coherence. Colour is speed $|\mathbf{u}|$, white curves are streamlines. A shallow slope spreads energy across scales (fine, ragged flow); a steep slope concentrates it in a few large scales (smooth, silky flow). Slope re-weights the *same* modes, so the large-scale skeleton persists while fine detail is added or removed.

2.1 Two dimensions: curl of a stream function

In two dimensions every divergence-free field is the curl of a scalar stream function ψ, $\mathbf{u} = (\partial_y \psi, -\partial_x \psi)$. Prescribing a vorticity spectrum $\hat{\omega}(\mathbf{k})$ and setting $\hat{\psi} = \hat{\omega}/|\mathbf{k}|^2$ gives

$$\hat{u}(\mathbf{k}) = +i\,\frac{k_y}{|\mathbf{k}|^2}\,\hat{\omega}(\mathbf{k}), \qquad \hat{v}(\mathbf{k}) = -i\,\frac{k_x}{|\mathbf{k}|^2}\,\hat{\omega}(\mathbf{k}), \tag{2}$$

with the sign convention $\omega = \partial_x v - \partial_y u = -\Delta\psi$, which satisfies $\mathbf{k} \cdot \hat{\mathbf{u}}(\mathbf{k}) = 0$ identically. There is no helicity in two dimensions; the only chirality available is the *sign* of the scalar vorticity, which biases whether vortices turn clockwise or counter-clockwise. We treat the genuine helicity control in three dimensions.

2.2 Three dimensions: the helical basis

The idea in one sentence: instead of describing each wave by its x-, y-, z-components, we describe it by how much it corkscrews clockwise versus counter-clockwise. These two corkscrew directions are the *helical* vectors below. They are special because the curl operator—the thing that measures local rotation—simply scales them rather than mixing them (they are its eigenvectors, (4)), so a swirl built from one of them reinforces its own rotation. Working in this basis makes handedness a single number and keeps the field divergence-free automatically.

For each $\mathbf{k} \neq \mathbf{0}$ choose a real orthonormal transverse frame $(\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2)$ with $\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2 \perp \mathbf{k}$ and $\{\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2, \hat{\mathbf{k}}\}$ right-handed ($\hat{\mathbf{k}} = \mathbf{k}/|\mathbf{k}|$). This is the Craya–Herring frame [7]. Define the two *helical* vectors [5, 6]

$$\mathbf{h}_\pm(\mathbf{k}) = \frac{\hat{\mathbf{e}}_1 \pm i\,\hat{\mathbf{e}}_2}{\sqrt{2}}. \tag{3}$$

They are transverse and are the eigenvectors of the curl operator in Fourier space:

$$\mathbf{h}_\pm \cdot \mathbf{k} = 0, \qquad i\,\mathbf{k} \times \mathbf{h}_\pm = \pm\,|\mathbf{k}|\,\mathbf{h}_\pm. \tag{4}$$

Any divergence-free field expands as $\hat{\mathbf{u}}(\mathbf{k}) = a_+(\mathbf{k})\mathbf{h}_+(\mathbf{k}) + a_-(\mathbf{k})\mathbf{h}_-(\mathbf{k})$, and the construction assigns the amplitudes $a_\pm$ their envelope and phase (§3–§5). Transversality of $\mathbf{h}_\pm$ guarantees $\operatorname{div}\mathbf{u} \equiv 0$ regardless of the amplitudes.

3 Helicity as a polarization control

In plain terms, this section builds the left/right corkscrew dial. Because the helical basis already separates clockwise swirls from counter-clockwise ones, biasing the handedness is just a matter of putting more energy into one than the other: a single number $p \in [-1, 1]$ slides from all-left through balanced to all-right. Everything below makes that precise.

The helicity of the field is diagonal in the helical basis. Writing $\boldsymbol{\omega} = \operatorname{curl}\mathbf{u}$ and using (4),

$$\mathcal{H} \;=\; \int_{\mathbb{T}^3} \mathbf{u} \cdot \boldsymbol{\omega}\, d\mathbf{x} \;=\; \sum_{\mathbf{k}} |\mathbf{k}|\big(\,|a_+(\mathbf{k})|^2 - |a_-(\mathbf{k})|^2\,\big). \tag{5}$$

We therefore split each mode's energy between the two channels by a single *helicity parameter* $p \in [-1, 1]$:

$$|a_+(\mathbf{k})|^2 = \tfrac{1+p}{2}\,E(\mathbf{k})^2, \qquad |a_-(\mathbf{k})|^2 = \tfrac{1-p}{2}\,E(\mathbf{k})^2, \tag{6}$$

so that p biases $\mathcal{H}$ directly. At $p = 0$ the two channels carry equal energy and the field is statistically mirror-symmetric ($\mathcal{H} = 0$ in the mean). At $p = \pm 1$ all energy is in one channel and the field is purely helical: $\boldsymbol{\omega} = \pm|\mathbf{k}|\,\mathbf{u}$ mode by mode, so $\boldsymbol{\omega} \parallel \mathbf{u}$ within each shell (a genuine Beltrami field only for a single shell; across shells the alignment is imperfect, §8). The ABC flows [9] are the classical single-shell example (Fig. 3).

A scale-independent readout of the effect is the *relative helicity*

$$\rho \;=\; \frac{\langle \mathbf{u} \cdot \boldsymbol{\omega} \rangle}{\sqrt{\langle |\mathbf{u}|^2 \rangle}\,\sqrt{\langle |\boldsymbol{\omega}|^2 \rangle}} \;\in [-1, 1], \tag{7}$$

with $\rho = \pm 1$ only for a single-shell Beltrami field. For a spectrum spanning many shells, $|\rho| < 1$ even at $p = \pm 1$, because $\mathbf{u}$ and $\boldsymbol{\omega}$ cannot be pointwise parallel across scales simultaneously; §8 measures this ceiling.

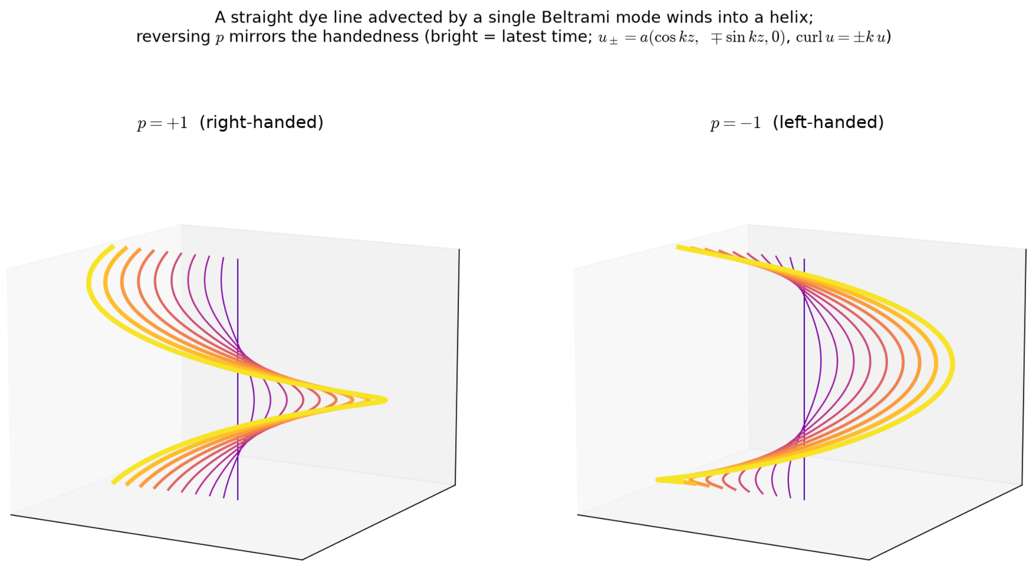

Figure 3: The chirality made concrete on a single Beltrami mode $\mathbf{u}(\mathbf{x}) = a(\cos kz, \mp \sin kz, 0)$, which satisfies $\mathrm{curl}\, \mathbf{u} = \pm k\, \mathbf{u}$ (the real-space, single-$\mathbf{k}$ realization of the helical basis of (3)). A straight dye line (dark) advected by the field winds into a one-turn helix (bright is the latest time); reversing the sign of p mirrors the handedness exactly. This is the single-shell limit in which $|\rho| = 1$; a multi-shell spectrum superposes many such windings, and the net $\int \mathbf{u} \cdot \boldsymbol{\omega}$ tracks p in sign and magnitude ($+8.5 \cdot 10^{-4}$ at $p = +1$, $-8.6 \cdot 10^{-4}$ at $p = -1$, ≈ 0 at $p = 0$ on a 32^3 grid).

4 Real fields and preservation of the helicity bias

Why this section matters to a practitioner: assembling the field from complex waves is convenient, but a velocity field you can actually advect through must be real-valued, and the standard step that makes it real pairs up opposite waves in a way that could, in principle, quietly cancel the left/right bias the artist just dialled in. It does not. The handedness survives that step exactly—the dial you set is the dial you get—and the rest of this section proves it. A reader who is content to take that on faith can skip to §5.

In detail: building modes with complex amplitudes yields a complex field; the standard remedy is to impose Hermitian symmetry $\hat{\mathbf{u}}(-\mathbf{k}) = \overline{\hat{\mathbf{u}}(\mathbf{k})}$, after which the inverse transform is exactly real. The nontrivial question is whether this symmetrization, which couples each $+\mathbf{k}$ mode to its $-\mathbf{k}$ partner, preserves the helicity bias (6) or averages it away. It preserves it.

Proposition 1. *Adopt the antipodal frame convention $\hat{\mathbf{e}}_1(-\mathbf{k}) = \hat{\mathbf{e}}_1(\mathbf{k})$, $\hat{\mathbf{e}}_2(-\mathbf{k}) = -\hat{\mathbf{e}}_2(\mathbf{k})$ (which keeps $\{\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2, \hat{\mathbf{k}}\}$ right-handed at $-\mathbf{k}$). Then*

$$\mathbf{h}_+(-\mathbf{k}) = \overline{\mathbf{h}_+(\mathbf{k})} = \mathbf{h}_-(\mathbf{k}), \tag{8}$$

and imposing $\hat{\mathbf{u}}(-\mathbf{k}) = \overline{\hat{\mathbf{u}}(\mathbf{k})}$ forces the helical amplitudes at $-\mathbf{k}$ to be $a_\pm(-\mathbf{k}) = \overline{a_\pm(\mathbf{k})}$. In particular $|a_+(-\mathbf{k})| = |a_+(\mathbf{k})|$: positive- and negative-helicity energy are each conserved, and the helicity of the realized real field is exactly $\mathcal{H} = 2\sum_{\mathbf{k}\in H} |\mathbf{k}|(|a_+(\mathbf{k})|^2 - |a_-(\mathbf{k})|^2)$ over a spectral half-space H (exact per realization, as the symmetrization conjugates only phases), i.e. the bias (6) survives verbatim.

The proof is a short two-line computation on the helical parity relation (8); it, together with the frame-convention and Nyquist-plane implementation notes, is deferred to Appendix A. The practical upshot is all that the construction needs: assemble $\hat{\mathbf{u}}(\mathbf{k})$ on a spectral half-space and set $\hat{\mathbf{u}}(-\mathbf{k}) = \overline{\hat{\mathbf{u}}(\mathbf{k})}$, and the handedness bias comes through untouched.

5 Phase coherence at fixed spectrum

This is the organization dial. Picture the same amount of motion at every scale—the flow is neither faster nor slower, finer nor coarser—but arranged two different ways: as an even, directionless haze, or gathered into a few distinct rolling vortices. What separates the two is not *how much* energy each scale carries but whether the waves at different scales line up in phase. This dial slides between those two arrangements while leaving the per-scale energy untouched, so the flow visibly reorganizes without changing pace. Curl-noise has no equivalent: its arrangement is whatever its potential happened to give.

The power spectrum $|\hat{\mathbf{u}}(\mathbf{k})|^2$ fixes how much energy sits at each scale; it says nothing about how the modes are *aligned* in phase. A field whose inter-mode phases are independent and uniform (the random-phase approximation, RPA) is Gaussian and spatially featureless; introducing correlations among the phases concentrates energy into localized coherent structures. That structural information lives in the phases rather than the amplitude spectrum is a standard observation [10].

We expose this as a control. Assign each helical amplitude its own unit-modulus phase,

$$a_\pm(\mathbf{k}) \;\longmapsto\; e^{i\varphi_\pm(\mathbf{k})}\, a_\pm(\mathbf{k}), \qquad \varphi_\pm(\mathbf{k}) = \theta^\pm_{\mathrm{ref}}(\mathbf{k}) + (1-\lambda)\,\theta_\pm(\mathbf{k}), \quad \lambda \in [0,1]. \tag{9}$$

Because the moduli $|a_\pm(\mathbf{k})|$ are untouched, the entire *helical energy spectrum*—the power spectrum $|a_+|^2 + |a_-|^2$ *and* the polarization split (6), hence the helicity—is frozen *exactly* at every λ; only the phases move. Here $\theta^\pm_{\mathrm{ref}}$ are structured reference phases (one per channel) and $\theta_\pm$ are independent uniform random phases, all drawn antisymmetric ($\theta(-\mathbf{k}) = -\theta(\mathbf{k})$) so the field stays real and (transversality being untouched) divergence-free. At $\lambda = 1$ each channel locks onto its reference; at $\lambda = 0$ they carry independent i.i.d. phases (an isotropic Gaussian haze, the RPA surrogate of the same amplitudes). Using *distinct* references for the two channels keeps the balanced field ($p = 0$) isotropic at every λ: a single shared reference would collapse a_+ and a_- to the same phase at $\lambda = 1$ and linearly polarize the $p = 0$ field. In two dimensions there is a single scalar amplitude, so (9) reduces to a common per-mode phase $e^{i(\theta_{\mathrm{ref}}+(1-\lambda)\theta)}$ on the stream function (used in E4). Sliding λ carries the field from formless noise to organized filaments *without changing the energy at any scale*—the axis that curl-noise lacks. (In the demonstrations of §8 the reference phases are those of a sparse set of localized vortex "blobs", so raising λ grows distinct rolling vortices out of the haze.) Figure 8 shows the two ends of this axis side by side at a fixed spectrum: the $\lambda = 0$ field is the featureless Gaussian haze that curl-noise produces, while $\lambda = 1$ pulls the same energy into distinct coherent vortices.

6 A real-time analytic variant

The two evaluation paths compute the same kind of field but serve opposite use cases, and the choice between them is a practical one. The FFT bake (§2) fills a full, dense spectrum on a grid once, then hands downstream animation a static texture—richest detail and exact tiling, but a per-rebuild cost, so it is the offline authoring tool. The mode-sum below sums a sparse handful of analytic waves and evaluates the field at any single point on demand, with no grid and no transform—cheaper to re-evaluate and resolution-independent, at the price of sparser detail and only approximate tiling, which is what makes it the real-time, in-shader option. They are not merely two implementations of one routine: the FFT path is dense and grid-baked, the mode-sum path is sparse and pointwise, and §8 (E3, E5) quantifies both the cost split and the one behavioural difference it introduces (a small residual handedness at $p = 0$). Use the FFT bake when authoring a hero field offline; use the mode-sum when the field must be evaluated live.

The FFT construction pays an N^d transform per rebuild—bake-scale, not real-time. An alternative sums a sparse set of analytic helical modes and evaluates the field *per point*, with no grid and no transform:

$$\mathbf{u}(\mathbf{x}) = \sum_{j=1}^{N} a_j \big(\hat{\mathbf{e}}_{1,j} \cos\phi_j - s_j\, \hat{\mathbf{e}}_{2,j} \sin\phi_j\big), \qquad \phi_j = \mathbf{k}_j \cdot \mathbf{x} + \theta_j, \tag{10}$$

with directions $\hat{\mathbf{k}}_j$ uniform on the sphere, amplitude $a_j = |\mathbf{k}_j|^{-s}$, transverse frame $(\hat{\mathbf{e}}_{1,j}, \hat{\mathbf{e}}_{2,j}) \perp \mathbf{k}_j$, and per-mode helicity sign $s_j \in \{+1, -1\}$ drawn with $\Pr(s_j = +1) = (1 + p)/2$. Because the directions are sampled uniformly on the sphere rather than on the integer lattice, the mode-sum does not reproduce the FFT field's exact per-shell energy; matching a target shell spectrum requires weighting the draws by the lattice mode density ($\propto |\mathbf{k}|^{d-1}$) or sampling $|\mathbf{k}_j|$ from it. This is one facet of the sparse-versus-dense trade-off (§9). Each term is the real part of a single helical mode: it is analytically transverse to its own $\mathbf{k}_j$, so $\operatorname{div}\mathbf{u} \equiv 0$ pointwise, and it is a Beltrami mode, $\operatorname{curl}\mathbf{u}_j = s_j|\mathbf{k}_j|\mathbf{u}_j$. Consequently the vorticity is a second cheap mode-sum requiring no derivatives, $\boldsymbol{\omega}(\mathbf{x}) = \sum_j s_j|\mathbf{k}_j|\big(\hat{\mathbf{e}}_{1,j} \cos\phi_j - s_j\hat{\mathbf{e}}_{2,j} \sin\phi_j\big)$, which makes (10) well suited to a fragment shader.

Two properties distinguish (10) from the FFT field. First, it is exactly divergence-free for every N (the only nonzero divergence one measures is finite-difference truncation, §8). Second—and this is a genuine caveat—at $p = 0$ a *finite* mode set does not cancel helicity exactly: the realized net helicity is a statistical fluctuation of size $\mathcal{O}(N^{-1/2})$ about zero, so the slider value p is only the *intended* bias. An implementation should therefore report the *measured* net $\mathbf{u} \cdot \boldsymbol{\omega}$ rather than p. The FFT construction does not share this caveat because it fills a full shell of modes.

A WebGL2 / GLSL ES 3.0 fragment shader implementing (10) directly accompanies the paper (`code/modesum.frag`): the M modes are uploaded once as a small $M \times 3$ float texture (rows $(\mathbf{k}, \theta)$, $(\hat{\mathbf{e}}_1, a)$, $(\hat{\mathbf{e}}_2, s)$, built exactly as `e3_mode_sum.py`), and each pixel runs the $\mathcal{O}(M)$ sum. A CPU port of the shader's exact arithmetic is divergence-free to $\max_j |\mathbf{k}_j \cdot \hat{\mathbf{e}}_{i,j}| = 5.6 \cdot 10^{-16}$ (the only nonzero divergence measured is $\mathcal{O}(h^2)$ finite-difference truncation), matching E3.

7 Solid obstacles

In plain terms: we want the flow to slide around a solid object—a rock, a character, a wall—without ever leaking into it, and without running a fluid solver to enforce that. The usual worry is that clamping a velocity field near a wall reintroduces the very compression the divergence-free construction was meant to avoid. The trick that avoids this is to damp the field's *potential* rather than the field itself, and only then take the curl: because the result is a curl, it is divergence-free by the same identity $\operatorname{div}\operatorname{curl} \equiv 0$ that underlies curl-noise, *for any obstacle and any damping profile.*

Concretely, every field of this paper has an analytic vector potential $\mathbf{A}$ ($\operatorname{curl}\mathbf{A} = \mathbf{u}$): on the helical basis each mode is a curl eigenvector (4), so $\mathbf{A} = \sum_{\mathbf{k}}(a_+\mathbf{h}_+ + a_-\mathbf{h}_-)/(\pm|\mathbf{k}|)$, and for the mode-sum (10) the potential is the term-by-term $\mathbf{A} = \sum_j \mathbf{u}_j/(s_j|\mathbf{k}_j|)$. Given a solid described by a signed-distance function $d(\mathbf{x})$ (positive outside, zero on the surface), we fade the potential to zero across a thin band of thickness θ with a smooth ramp α of the distance and take the curl of the faded potential:

$$\mathbf{u}_b(\mathbf{x}) \;=\; \mathrm{curl}\big(\alpha(d/\theta)\,\mathbf{A}\big) \;=\; \tfrac{1}{\theta}\,\alpha'(d/\theta)\,\big(\nabla d \times \mathbf{A}\big) \;+\; \alpha(d/\theta)\,\mathbf{u}, \tag{11}$$

using $\mathrm{curl}(f\mathbf{A}) = \nabla f \times \mathbf{A} + f\ \mathrm{curl}\,\mathbf{A}$. The ramp is the Bridson curl-noise boundary quintic [1],

$$\alpha(q) = \tfrac{1}{8}\,q\,(15 - 10q^2 + 3q^4) \ \text{ on } [0,1], \qquad \alpha(q \le 0) = 0,\ \ \alpha(q \ge 1) = 1, \tag{12}$$

with $\alpha(0) = 0$ but $\alpha'(0) = \frac{15}{8} > 0$, so the wall value is a pure tangential slip flow rather than a no-slip dead zone, and $\alpha(1) = 1$ with $\alpha'(1) = \alpha''(1) = 0$, so the field blends C^2-smoothly back into the unconstrained one at the edge of the band (Fig. 4). Inside the solid ($d \le 0$) the field is identically zero.

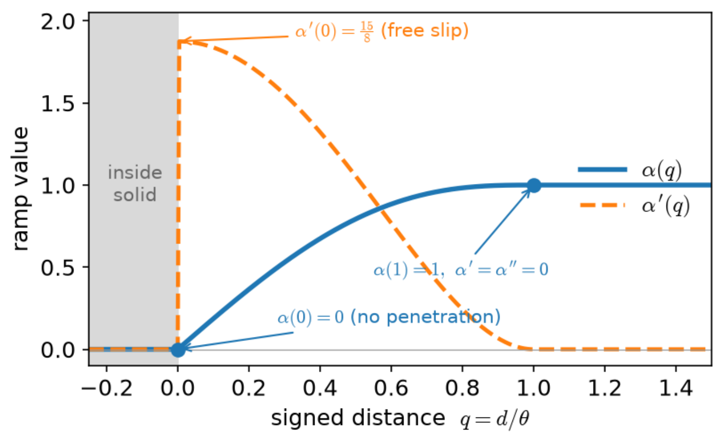

Figure 4: The Bridson free-slip quintic (12) used to fade the vector potential across the obstacle band. The ramp α (solid) rises from 0 at the wall to 1 at the band edge; its derivative α' (dashed) is nonzero at the wall ($\alpha'(0) = \frac{15}{8}$, giving tangential slip rather than a dead zone) and vanishes to second order at the edge, so the bounded field meets the unconstrained one C^2-smoothly. The distance is normalized by the band thickness, $q = d/\theta$.

Two properties hold *by construction*, wherever the signed distance is smooth (so that ∇d exists—away from the medial axis and any edges or corners of the obstacle, a measure-zero set the ramp never needs to resolve). First, $\mathbf{u}_b$ is a curl, so it is exactly divergence-free: classically outside the surface, and distributionally across it (the field is zero inside $d < 0$ and a tangential slip flow just outside, so there is a velocity discontinuity at $d = 0$, as for any free-slip wall—see below). Second, it is tangent to the surface: on $d = 0$ the first term of (11) carries the factor $\nabla d \times \mathbf{A}$, whose component along the surface normal $\hat{\mathbf{n}} = \nabla d$ is $(\nabla d \times \mathbf{A}) \cdot \nabla d = 0$ identically, while the second term carries the factor $\alpha(0) = 0$; hence $\mathbf{u}_b \cdot \hat{\mathbf{n}} = 0$ on the wall. Because the wall normal velocity vanishes identically, $\{d = 0\}$ is an invariant surface of the exact flow: a continuous trajectory cannot cross it, so containment is a theorem for the continuous field and only the time integrator can leak (E6 measures that leak). This is precisely the free-slip, boundary-respecting behaviour targeted by pointwise incompressible interpolation for grid-based fluids [13], obtained here in closed form on the analytic field with no grid and no projection. The ramp α is C^2 at the band edge $q = 1$ but only one-sided at the wall ($\alpha'(0^-) = 0 \neq \frac{15}{8} = \alpha'(0^+)$), which is exactly the slip discontinuity noted above. Beyond the band ($d \ge \theta$) the field is bit-identical to the unconstrained one.

(E6) Obstacle: tangency, containment, and machine-zero divergence (`e6_obstacle.py`). We place a sphere of radius $R = 1$ in the mode-sum field ($N = 40$ modes, slope $s = 1$, band $\theta = 0.6$, pinned seed) and measure the three guarantees (Fig. 5). **(a) Tangency:** over 20,000 surface points the peak normal flux relative to the rms speed is $\max |\mathbf{u}_b \cdot \hat{\mathbf{n}}|/\mathrm{rms}|\mathbf{u}_b| = 4.8 \cdot 10^{-16}$—machine zero, the free-slip guarantee holding exactly. **(b) Containment:** advecting 5000 tracer particles seeded in and around the influence band through the steady field with RK4, the fraction that end up *inside* the sphere is 0.00% with the ramp against 2.84% without it (raw field), matching the independent GPU particle-readback figure of 0 versus ≈2.9%. **(c) Divergence:** $\mathbf{u}_b$ is an analytic curl, so $\operatorname{div} \mathbf{u}_b = 0$ identically; symbolic algebra confirms $\operatorname{div} \operatorname{curl}(f\mathbf{A}) = 0$, and a complex-step evaluation of the analytic divergence in the band interior gives rms $1.2 \cdot 10^{-15}$ (machine zero), whereas a naive central-difference stencil at $h = 0.01$ reports rms $7.2 \cdot 10^{-4}$—the $\mathcal{O}(h^2)$ truncation of the stencil, not an error in the field. Every number here is emitted by `e6_obstacle.py`.

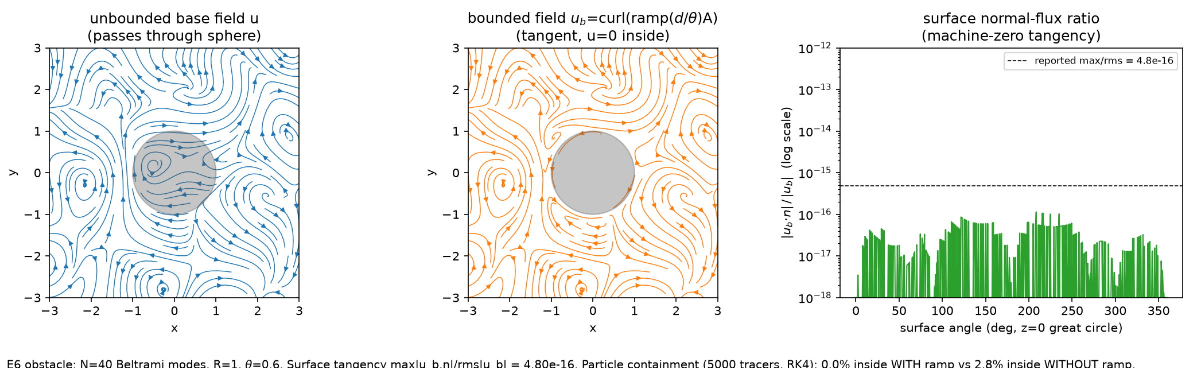

Figure 5: Solid-obstacle handling by ramping the vector potential before the curl (11). **Left:** the unconstrained field passes straight through the sphere. **Middle:** the bounded field $\mathbf{u}_b = \mathrm{curl}(\alpha(d/\theta)\mathbf{A})$ bends tangentially around it and is zero inside—exactly divergence-free because it is a curl, and free-slip because $\alpha(0) = 0$. **Right:** the surface normal-flux ratio $|\mathbf{u}_b \cdot \hat{\mathbf{n}}|/|\mathbf{u}_b|$ at points around the sphere sits at machine zero ($\max/\mathrm{rms} = 4.8 \cdot 10^{-16}$). Of 5000 advected particles, 0.00% finish inside the sphere with the ramp versus 2.84% without it (`e6_obstacle.py`, sphere $R = 1$, $N = 40$ modes, band $\theta = 0.6$).

8 Numerical validation

We report six experiments (E1–E5 and E7), each reproduced by a self-contained script accompanying the paper (`code/e1_*..e5_*`, `code/e7_*`); the obstacle experiment E6 is reported in §7 (`code/e6_obstacle.py`). E4 validates the phase-coherence axis on the 2D construction and E7 on the 3D helical builder that also generates the figures. The printed numbers below are exactly those the scripts emit at the pinned seed.

(E1) Helical identities (`e1_helical_identities.py`). Over 2000 random real wavevectors, the transversality and curl-eigenvector relations (4) hold to $\|\mathbf{h}_\pm \cdot \mathbf{k}\| \le 4.1 \cdot 10^{-16}$ and $\|i\mathbf{k} \times \mathbf{h}_\pm \mp |\mathbf{k}|\mathbf{h}_\pm\| \le 1.0 \cdot 10^{-15}$; in symbolic algebra both are identically zero. The parity identity (8) holds *exactly* (residual 0), and the amplitude map of Proposition 1 to $2.4 \cdot 10^{-15}$ numerically and exactly in symbolic algebra.

(E2) FFT field: real, divergence-free, helicity linear in p (`e2_fft_field.py`). On a 32^3 grid with slope $s = 1.6$, enforcing Hermitian symmetry:

p	$\max \lvert\mathbf{k}\cdot\hat{\mathbf{u}}\rvert/\lvert\hat{\mathbf{u}}\rvert$	$\max \lvert\mathrm{Im}\,\mathbf{u}\rvert$	ρ
-1.0	$4.4 \cdot 10^{-17}$	$4.7 \cdot 10^{-20}$	-0.7961
-0.5	$6.3 \cdot 10^{-17}$	$4.9 \cdot 10^{-20}$	-0.3981
0.0	$5.8 \cdot 10^{-17}$	$5.3 \cdot 10^{-20}$	0.0000
0.5	$5.8 \cdot 10^{-17}$	$4.9 \cdot 10^{-20}$	0.3981
1.0	$4.4 \cdot 10^{-17}$	$5.4 \cdot 10^{-20}$	0.7961

The field is divergence-free and real to machine precision, and ρ is linear in p (fit $\rho = 0.79612\,p$, maximum deviation from the line $1.1 \cdot 10^{-16}$), zero at $p = 0$, and correct in sign. The ceiling $|\rho(\pm 1)| \approx 0.80$ is below 1 because the spectrum spans many shells; it is spectrum-dependent ($\rho(+1) = 0.821$, 0.796, 0.812 at slopes $s = 1.2$, 1.6, 2.0), consistent with the single-shell remark after (7).

(E3) Mode-sum: exact transversality and the $N^{-1/2}$ helicity residual (`e3_mode_sum.py`). For (10) the analytic transversality is machine-exact, $\max_j |\mathbf{k}_j \cdot \hat{\mathbf{e}}_{i,j}| = 4.4 \cdot 10^{-16}$, so $\operatorname{div} \mathbf{u} = 0$ analytically; the central-difference divergence falls cleanly as $\mathcal{O}(h^2)$ (rms $2.6 \cdot 10^{-2} \to 4.1 \cdot 10^{-4}$ as h halves from 0.1, Richardson ratios $3.96, 3.99, 4.00$), confirming truncation rather than construction error. With $N = 40$ modes, $\rho(p) = (-0.889, -0.662, +0.069, +0.383, +0.888)$ for $p = (-1, -\frac{1}{2}, 0, \frac{1}{2}, 1)$, monotone and saturating near ± 1; the $+0.069$ at $p = 0$ is one draw's grid-measured relative helicity, a realization of the finite-N fluctuation. To exhibit its decay we use the $|\mathbf{k}|$-weighted net-helicity estimator $h_N := \big(\sum_j s_j |\mathbf{k}_j| a_j^2\big) / \big(\sum_j |\mathbf{k}_j| a_j^2\big)$, whose mean is p; averaged over 200 draws at $p = 0$,

$$\mathrm{rms}\, h_N \;=\; 0.253,\ 0.176,\ 0.114,\ 0.090,\ 0.064 \quad \text{for } N = 20, 40, 80, 160, 320,$$

i.e. $\mathrm{rms}\, h_N \cdot \sqrt{N} \approx 1.0$–$1.14$ (fitted exponent -0.494 vs. the predicted $-\frac{1}{2}$), the $\mathcal{O}(N^{-1/2})$ fluctuation of §6. (The grid-measured ρ and h_N are two $\mathcal{O}(N^{-1/2})$ proxies for the same finite-N net helicity.) Figure 6 plots both results.

(E4) Phase coherence freezes the spectrum while changing structure (`e4_phase_coherence.py`). The phase-coherence axis (§5) claims to move the field from formless noise to organized filaments *without touching the energy at any scale.* We verify both halves on the 2D construction (160^2 grid, slope $s = 1.4$). As the coherence knob λ is swept $0 \to 1$ under the phase interpolation (9), the total spectral energy is invariant to $\sum |\hat{\mathbf{u}}|^2 = 5.209353$ at every λ (spread $8.9 \cdot 10^{-16}$, relative $1.7 \cdot 10^{-16}$), and the per-shell energy spectrum is likewise frozen (maximum relative deviation of any shell across λ is $1.8 \cdot 10^{-16}$); the field stays divergence-free ($\max |\mathbf{k} \cdot \hat{\mathbf{u}}|/|\hat{\mathbf{u}}| \le 1.4 \cdot 10^{-17}$) and real ($\max |\mathrm{Im}\,\mathbf{u}| \sim 10^{-7}$—the phase-extraction floor of the reference blobs, where arg of a near-null mode is ill-conditioned, not machine ϵ; the advected field is its real part regardless). The spatial organization, by contrast, changes monotonically: the flatness (kurtosis) of the vorticity, which is ≈ 3 for a random-phase field and grows as energy concentrates into sparse coherent structures, rises

$$\text{flatness}(\omega) \;=\; 2.75,\ 2.88,\ 3.05,\ 5.35,\ 9.22 \qquad \text{for } \lambda = 0,\ 0.25,\ 0.5,\ 0.75,\ 1$$

(16-draw mean; the $\lambda = 0$ value sits just below 3 because the fixed-amplitude modes make the random-phase surrogate mildly sub-Gaussian), from a near-Gaussian haze at $\lambda = 0$ up to strongly intermittent at $\lambda = 1$. Structure appears; energy per scale does not move.

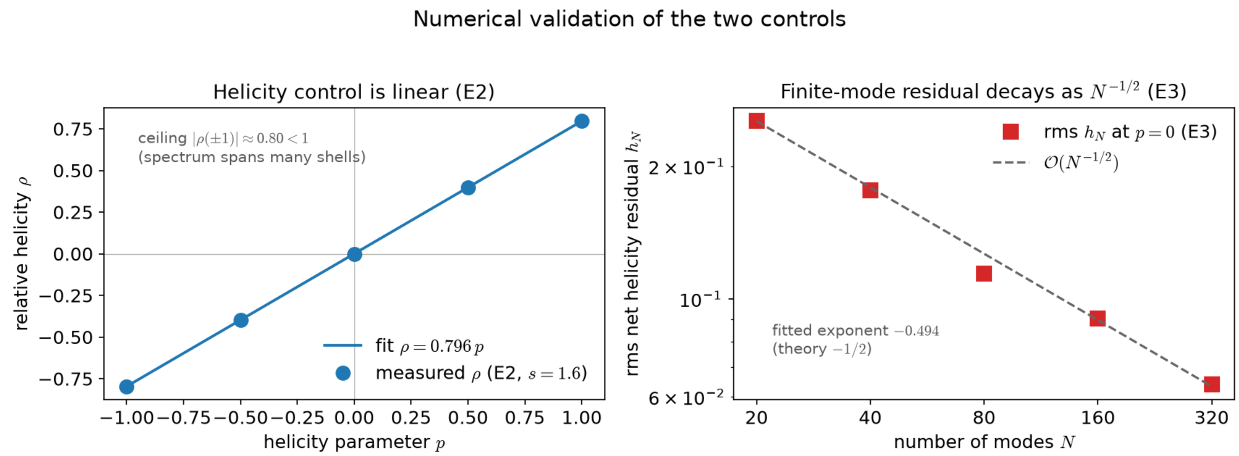

Figure 6: The two quantitative results of E2 and E3. **Left:** for the FFT field the relative helicity ρ is linear in the knob p (fit $\rho = 0.796\,p$); the ceiling $|\rho(\pm 1)| \approx 0.80 < 1$ reflects a spectrum spanning many shells (7). **Right:** for the mode-sum, the net-helicity residual at $p = 0$ decays as $\mathcal{O}(N^{-1/2})$ with mode count (fitted exponent -0.494), the finite-N fluctuation of §6.

(E7) The same, in 3D, on the figure-generating builder (`e7_phase_coherence_3d.py`). E4 exercises the 2D stream-function path; E7 runs the identical test on the 3D helical builder that generates the teaser and advection figures (`_helical.build_fft_3d`), so the figure code is itself under test. On a 48^3 grid ($s = 1.4$, $p = 0.6$), sweeping $\lambda = 0 \to 1$ under (9) holds the total energy fixed to $\sum |\hat{\mathbf{u}}|^2 = 36.059$ *exactly* (relative spread $2.0 \cdot 10^{-16}$), holds the net helicity fixed to $\mathcal{H} = 7.85 \cdot 10^{-4}$ (relative spread $1.4 \cdot 10^{-16}$) and the per-shell spectrum fixed (max deviation $1.7 \cdot 10^{-16}$), and keeps the field divergence-free ($\leq 9 \cdot 10^{-17}$) and real ($\sim 4 \cdot 10^{-20}$). The spatial organization again grows monotonically—the vorticity-magnitude flatness (the $|\boldsymbol{\omega}|$ analogue of the E4 statistic; ≈ 3.1 is the Gaussian baseline for the magnitude of three components) rises

$$\text{flatness}(|\boldsymbol{\omega}|) \;=\; 3.11,\ 3.33,\ 10.88,\ 23.35,\ 26.70 \qquad \text{for } \lambda = 0,\ 0.25,\ 0.5,\ 0.75,\ 1$$

(8-draw mean)—while the field's relative helicity at $p = +1$ is $\rho \approx 0.77$, matching the FFT construction of E2, and the balanced field ($p = 0$) stays isotropic at every λ. This is the check whose absence let an earlier 3D builder ship with the coherence knob inoperative (it applied a common phase on top of already-random amplitudes, so flatness($|\boldsymbol{\omega}|$) stayed flat at ≈ 3.1 for every λ); the present builder (independent per-channel phases, each locking onto its own structured reference, (9)) makes coherence respond to λ while freezing spectrum and helicity.

(E5) The two cost regimes (`e5_performance.py`). The two constructions occupy different cost budgets, whose scaling Fig. 7 measures (single-machine CPU `numpy` reference; the scaling is the claim, absolute times are machine-dependent). The FFT path is a *bake* cost—one $\mathcal{O}(N^3 \log N)$ rebuild per authored field, after which the field is a static texture (3.0 MB of `float32 vec3` at 64^3) and advection downstream carries no solver cost. The mode-sum path trades the grid for an $\mathcal{O}(M)$ per-point evaluation whose throughput falls as $1/M$; the arithmetic is embarrassingly parallel across pixels, which is why the fragment-shader implementation (`code/modesum.frag`), not a scalar CPU loop, is the real-time target.

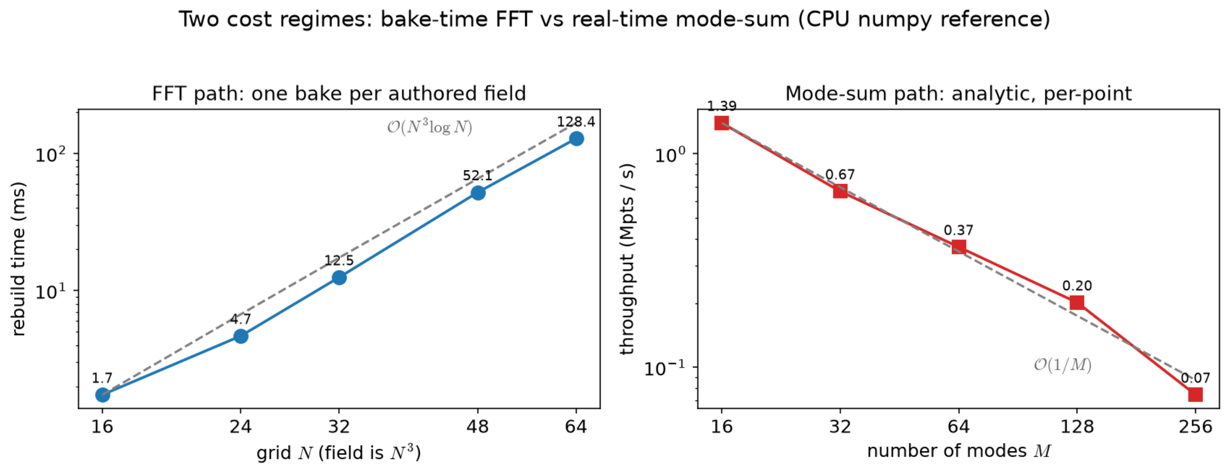

Figure 7: Two cost regimes (CPU `numpy` reference, one machine, no GPU). **Left:** the FFT construction rebuilds an N^3 field in $\mathcal{O}(N^3 \log N)$; a bake-time cost paid once per authored field. **Right:** the analytic mode-sum evaluates per point at $\mathcal{O}(M)$ for M modes, so throughput scales as $1/M$. The dashed lines are the predicted scalings. Absolute mode-sum rates are for a single CPU core; the per-pixel arithmetic parallelises directly on a GPU (`code/modesum.frag`).

9 Discussion and limitations

Relation to curl-noise. Curl-noise [1] takes a noise potential in *real* space and returns its curl, divergence-free by the identity $\operatorname{div}\operatorname{curl} \equiv 0$; we assemble the field in the *frequency* domain on the helical basis, divergence-free by transversality $\mathbf{h}_\pm \cdot \mathbf{k} = 0$. The two mechanisms differ in what they leave free to control (Table 1): curl-noise is fixed once the potential is chosen—its handedness is not a parameter, and its phases are those of the potential—whereas the helical representation exposes the $\pm$ polarization split (helicity, §3) and the common per-mode phase (coherence, §5) as continuous knobs. Our $\lambda = 0$ field sits at the random-phase, mirror-symmetric, isotropic corner of this space, which is the same corner curl-noise occupies—both are featureless at a fixed spectrum (equal-spectrum vorticity flatness ≈ 3.0, the Gaussian value, versus ≈ 11.9 at $\lambda = 1$; Fig. 8). We stress that this is a comparison against a *random-phase spectral surrogate* built at the matched spectrum, not against a specific curl-of-Perlin implementation, whose lattice and octave structure imposes its own weak phase correlations; the point is only that the featureless end of the coherence axis is curl-noise-*like*, and that λ opens an axis curl-noise leaves fixed. The FFT path shares curl-noise's per-frame cost class but not its constant-time *evaluation*; the mode-sum variant (§6) is the real-time offering.

Relation to incompressible solvers. A separate line of graphics work builds divergence-free fields with explicit vortex and helicity structure inside a time-stepping *simulator*: Schrödinger's smoke [11] evolves an incompressible field from a wavefunction, and implicit vortex filaments [12] carry controllable vortex topology; procedural detail is likewise added on top of solvers by guided and stream-guided synthesis [4, 17] (see [18] for a recent survey). The present method is complementary: it is an authoring generator of a static field with no solver and no time integration, exposing helicity as a continuous polarization knob rather than as an emergent property of dynamics.

Limits.

	Curl-noise [1]	**This work**
Built in	real space (noise potential)	frequency domain (helical basis)
Div-free because	div curl $\equiv 0$	$\mathbf{h}_\pm \cdot \mathbf{k} = 0$ (transversality)
Spectral slope	yes (potential band)	yes ($E(\mathbf{k})$, (1))
Helicity / chirality	not a control	knob $p \in [-1, 1]$ (§3)
Phase organization	tied to potential	knob $\lambda \in [0, 1]$ at fixed spectrum
Evaluation	$\mathcal{O}(1)$ per point	$\mathcal{O}(N^d \log N)$ rebuild (FFT) or $\mathcal{O}(M)$ per point (mode-sum)

Table 1: Curl-noise and the present construction both produce exactly divergence-free fields, but by different mechanisms, which is what makes helicity and phase organization controllable here and not there. The random-phase, mirror-symmetric, isotropic setting of our method occupies the same featureless corner as curl-noise (Fig. 8).

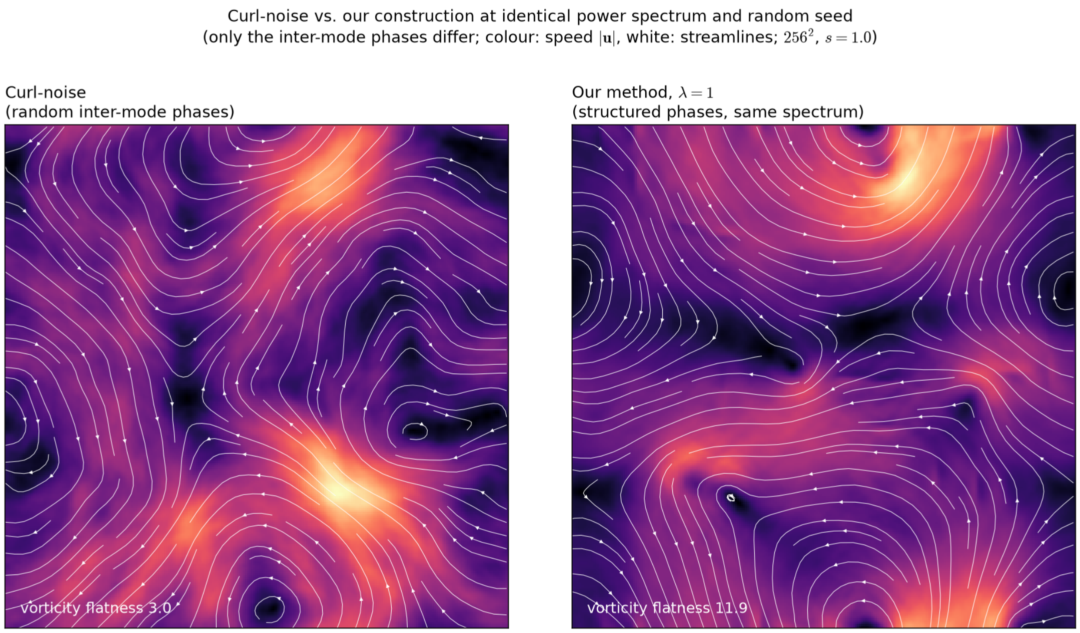

Figure 8: The random-phase corner ($\lambda = 0$, left; the curl-noise-like, spectrally matched surrogate—see text) and our construction at $\lambda = 1$ (right), built on the *same* power spectrum and the same random seed, so that only the inter-mode phases differ. Colour is speed $|\mathbf{u}|$, white curves are streamlines. The random phases give a homogeneous field (vorticity flatness ≈ 3.0, the Gaussian value); the structured phases concentrate the same energy into distinct coherent vortices (vorticity flatness ≈ 11.9) without changing the spectrum—the organization axis curl-noise does not expose. (Speed flatness, by contrast, barely moves between the two panels; the reorganization shows in the vorticity, as in E4.)

- The FFT path costs an N^d transform per rebuild and is an authoring/bake tool; the practical deployment bakes the field to a texture or an OpenVDB volume and advects downstream.
- The mode-sum path is real-time but sparse: it trades away exact tiling and dense-shell detail, and its $p = 0$ state is only mirror-symmetric in the mean ($\mathcal{O}(N^{-1/2})$ residual, §6).
- The relative-helicity ceiling $|\rho(\pm 1)| < 1$ is intrinsic to a multi-shell spectrum; a single-shell field reaches ± 1 but looks monochromatic.

Applications. The fields drive the standard procedural pipelines directly (Fig. 9): particle advection, semi-Lagrangian smoke density, and volume raymarching of an advected scalar. Because the field is divergence-free by construction, this transport needs no pressure projection—the expensive step of a fluid solver—so the two new controls add no solver cost to the downstream renderer (the advection itself, of course, still costs what advection costs).

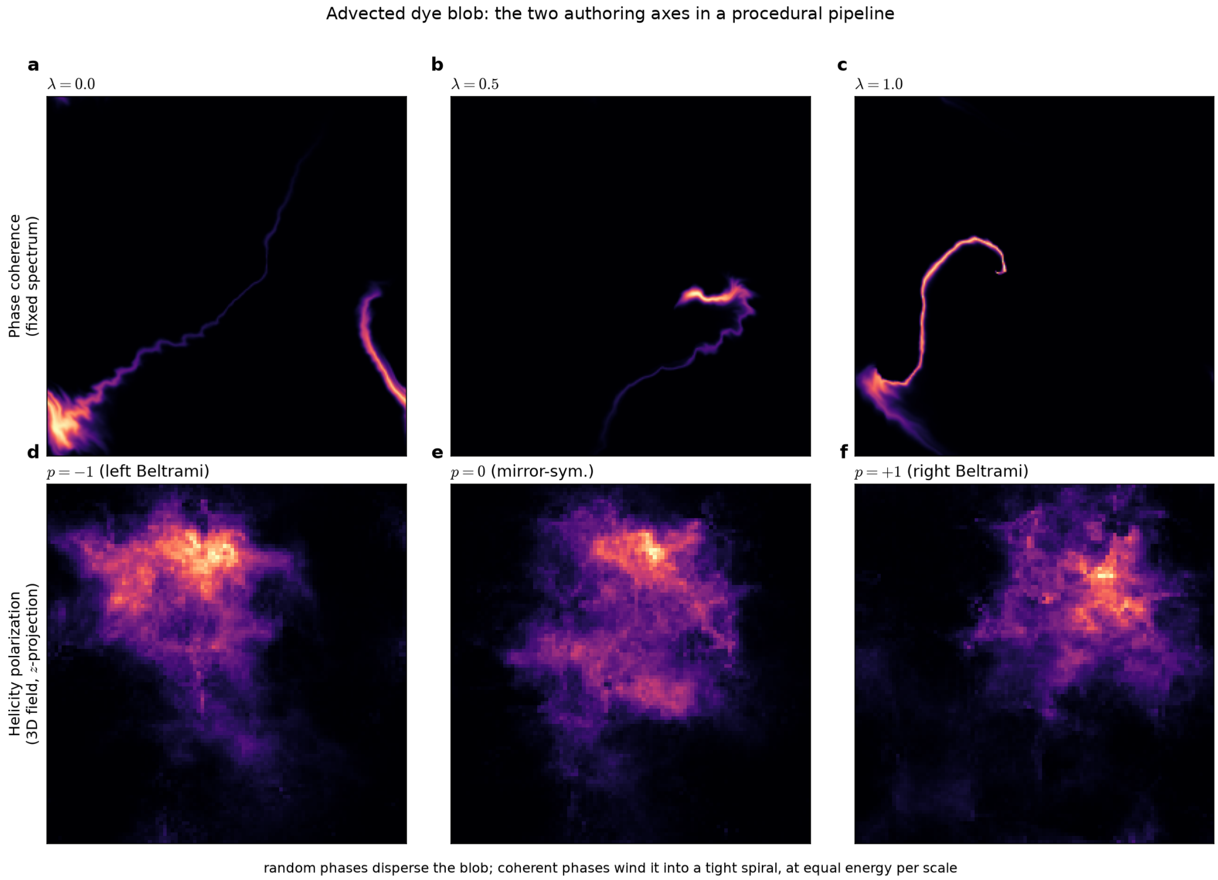

Figure 9: Both axes drive the standard procedural pipeline. A compact dye blob is transported by semi-Lagrangian advection through the steady field; brightness is the advected dye. **Top:** the phase-coherence knob—at $\lambda = 0$ the blob disperses into scattered streaks, while $\lambda = 1$ winds it into a single tight coherent spiral. **Bottom:** the helicity polarization p. The transport needs no pressure projection, the field being divergence-free by construction.

Conclusion. Building divergence-free fields on the helical basis gives the artist three independent dials—the spectral slope, the chirality (helicity), and the degree of organization at fixed spectrum (phase coherence)—the latter two otherwise inaccessible to curl-noise, all continuously variable and all keeping divergence-freeness exact. The same potential-ramp mechanism handles solid obstacles with a measured free-slip, no-penetration guarantee and no solver (§7). The constructions are elementary and reproducible.

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Code and data availability. All code and data are available at `https://github.com/rifmj/helical-fields` (DOI: 10.5281/zenodo.21246333). Every number reported in this paper is regenerated by a pinned-seed script via `make reproduce`, and every figure is regenerated from source via `make figures`.

A Proof of Proposition 1 and implementation notes

Proof of Proposition 1. With $\hat{\mathbf{k}}(-\mathbf{k}) = -\hat{\mathbf{k}}(\mathbf{k})$ and the stated frame convention, $\mathbf{h}_+(-\mathbf{k}) = \frac{1}{\sqrt{2}}(\hat{\mathbf{e}}_1(-\mathbf{k}) + i\,\hat{\mathbf{e}}_2(-\mathbf{k})) = \frac{1}{\sqrt{2}}(\hat{\mathbf{e}}_1(\mathbf{k}) - i\,\hat{\mathbf{e}}_2(\mathbf{k})) = \overline{\mathbf{h}_+(\mathbf{k})} = \mathbf{h}_-(\mathbf{k})$, which is (8). Write $\hat{\mathbf{u}}(\mathbf{k}) = a_+\mathbf{h}_+(\mathbf{k}) + a_-\mathbf{h}_-(\mathbf{k})$. Then $\overline{\hat{\mathbf{u}}(\mathbf{k})} = \overline{a_+}\,\overline{\mathbf{h}_+(\mathbf{k})} + \overline{a_-}\,\overline{\mathbf{h}_-(\mathbf{k})} = \overline{a_+}\,\mathbf{h}_-(\mathbf{k}) + \overline{a_-}\,\mathbf{h}_+(\mathbf{k})$. Expanding $\hat{\mathbf{u}}(-\mathbf{k}) = b_+\mathbf{h}_+(-\mathbf{k}) + b_-\mathbf{h}_-(-\mathbf{k}) = b_+\mathbf{h}_-(\mathbf{k}) + b_-\mathbf{h}_+(\mathbf{k})$ and matching to $\overline{\hat{\mathbf{u}}(\mathbf{k})}$ gives $b_+ = \overline{a_+}$, $b_- = \overline{a_-}$. Hence $|a_+(-\mathbf{k})| = |a_+(\mathbf{k})|$ and likewise for a_-; substituting into (5) and pairing $\pm\mathbf{k}$ gives the stated half-space sum. The symmetrization conjugates phases; it moves no energy between the $\pm$ channels. □

Remark 1 (convention). The conclusion depends on the antipodal frame choice $\hat{\mathbf{e}}_2(-\mathbf{k}) = -\hat{\mathbf{e}}_2(\mathbf{k})$; the opposite choice $\hat{\mathbf{e}}_2(-\mathbf{k}) = +\hat{\mathbf{e}}_2(\mathbf{k})$ would instead swap the $\pm$ channels between $\mathbf{k}$ and $-\mathbf{k}$. In practice one need not track the frame at $-\mathbf{k}$ at all: assembling $\hat{\mathbf{u}}(\mathbf{k})$ on a spectral half-space and setting $\hat{\mathbf{u}}(-\mathbf{k}) = \overline{\hat{\mathbf{u}}(\mathbf{k})}$ directly realizes the preserving convention.

Remark 2 (Nyquist plane). On an N^d grid the Nyquist frequencies ($k_i = -N/2$ on some axis) are self-conjugate under the discrete Fourier modular arithmetic and have no distinct $-\mathbf{k}$ partner; naïve mirror-conjugation there leaks divergence. The fix is to zero the Nyquist plane. With the cutoff $k_c \ll N/2$ of (1) these modes carry negligible energy, but a robust implementation zeroes them explicitly.

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